

46/50

92%

ABERDARE TECHNICAL COLLEGE
INFORMATION COMMUNICATION
TECHNOLOGY

date 18/09/2022

ELEMENT: DESIGN COMPUTER NETWORK

UNIT CODE: IT/CU/ICTA/CR/02/5/MA

UNIT NAME: NETWORK DESIGN & MANAGEMENT

LEVEL 5, MODULE 3

TERM 3

TOTAL MARKS: 50 MARKS

Name.....Marvin Ndugire.....Date.....

Sign.....[Signature].....Cdacc regNo.....

Instructions:

Answer all questions.

1. Define a computer network and state two benefits of computer networking. (4 marks)
2. State and explain four types of computer networks based on geographical coverage. (8 marks)
3. Explain the following network topologies:
 - a) Star
 - b) Ring
 - c) Mesh
 - d) Hybrid
 - e) Point-to-point (10 marks)
4. State four components of a computer network and explain the function of each. (8 marks)
5. State three examples of network transmission media and give one characteristic of each. (6 marks)
6. Explain the following stages involved in collecting and handling computer network user requirements:
 - a) User requirements identification

1 A Computer Network is a group of Computers and devices connected together to share information and resources.

Sharing of files and information.

Sharing of resources such as printers and internet.

2 PAN (Personal Area Network) - A network that covers a very small area around one person.

LAN (Local Area Network) - A network that covers a smaller area such as a home, school or office.

MAN (Metropolitan area network) - A network that covers a larger area such as a town.

Wide area network - A network that covers a very large geographical area.

3 Star - all devices are connected to a central device such as a switch.

Ring - Each computer is connected to two other computers, forming a circle.

Mesh - Devices are connected to several or all other devices, providing alternative paths for data.

Hybrid - A combination of two or more different network topologies.

Point-to-Point - Two devices are directly connected to communicate with each other.

Router - Connects different networks and directs data between them.

Switch - Connects devices within a network and forwards data to the correct device.

Network interface card - Allows a computer to connect to a network.

Modem - Converts signals to allow communication over an internet connection.

5 Twisted-pair cable - Cheap and commonly used in LANs.

Coaxial cable - Has good resistance to interference.

Fibre-optic cable - Transmits data at very high speeds using light.

6 User Requirements Identification

It involves finding out what users need from the network by interviewing users, observing their work and collecting their suggestions and problems.

7 User Requirements Analysis

This is the process of finding out and understanding what users need from a computer system or network before it is designed or installed.

User Requirements Documentation

It involves writing down the users' needs and requirements clearly for reference during system or network design.

Physical network design - Shows the actual location and connection of network devices, cables, computers and other hardware.

Logical network design - Shows how data moves through the network, including IP addresses, protocols and network segments.

8 Greenfield site - A new network is designed and installed from scratch where there is no existing network infrastructure.

Brownfield site - A network is designed or upgraded using existing network infrastructure.

9 Security
Reliability